

Physics Mock Test

Magnetism and Matter

Class 12 | UNIT 3 (B)

Marks: 24

Time: 60 Minutes

Syllabus: Magnetic dipole moment, Earth's magnetism, property of magnetic materials (Dia, Para, Ferro), Hysteresis, and Magnetization.

Instructions: For Section 2, state the formulas used clearly. Assumed value: $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$

Section 1: Multiple Choice Questions (1 Mark Each)

1. Bar Magnet Polarity

The magnetic dipole moment of a bar magnet is a vector quantity defined as being directed from:

- a) North to South pole
- b) South to North pole
- c) Geographic North to South
- d) Center to either pole

2. Earth's Magnetic Elements

The angle between the magnetic meridian and the geographic meridian at a specific place on Earth is known as:

- a) Angle of Dip
- b) Angle of Declination
- c) Angle of Inclination
- d) Horizon Angle

3. Diamagnetic Specimen

Identify the diamagnetic material among the list provided below:

- a) Iron
- b) Nickel
- c) Copper
- d) Cobalt

4. Magnetization defined

The Intensity of Magnetization (M) within a material is most accurately defined as:

- a) Magnetic moment per unit volume
- b) Magnetic field per unit area
- c) Magnetic flux per unit current
- d) Induced EMF per unit time

5. Paramagnetic Susceptibility

At room temperature, typical paramagnetic materials exhibit a magnetic susceptibility (χ) that is:

- a) Negative and very small
- b) Positive and small
- c) Positive and very large
- d) Zero

6. Hysteresis Loop Efficiency

To minimize energy loss (hysteresis loss) in the core of a transformer, the chosen material should possess:

- a) A wide hysteresis loop
- b) A narrow hysteresis loop
- c) High coercivity
- d) High retentivity only

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Pole Strength Calculation

A bar magnet has a magnetic dipole moment of $0.5 \text{ A}\cdot\text{m}^2$. If the physical length of the magnet is 10 cm, calculate its magnetic pole strength.

8. Earth's Field Components

At a certain location, the angle of dip is 30° . Calculate the ratio of the horizontal component (B_H) to the vertical component (B_V) of the Earth's magnetic field at that place.

9. Magnetization of a Core

A solenoid with 1000 turns per meter carries a current of 2 A. When a material is placed inside, the measured magnetic field is $4\pi \times 10^{-3} \text{ T}$. Find the magnetization (M) of the core material.

10. Susceptibility and Permeability

For a certain magnetic material, the relative permeability (μ_r) is found to be 1.5. Calculate the magnetic susceptibility (χ) of this material and identify the class of magnetic material it belongs to.

11. Hysteresis Terminology

Define the terms **Retentivity** and **Coercivity**. How do these properties dictate the choice of materials for making permanent magnets versus electromagnets?

12. Comparative Magnetism

Distinguish between paramagnetic and ferromagnetic materials based on their behavior in a non-uniform magnetic field. Give one example of each.

Answer Key & Solutions

1. b) South to North pole

By convention, the dipole moment vector points from S to N.

2. b) Angle of Declination

Declination is the angle between geographic and magnetic meridians.

3. c) Copper

Copper is weakly repelled by magnets (diamagnetic). Iron, Nickel, and Cobalt are ferromagnetic.

4. a) Magnetic moment per unit volume

$M = m_{\text{net}} / V$. Units are A/m.

5. b) Positive and small

Paramagnetic substances align weakly with the field.

6. b) A narrow hysteresis loop

The area of the loop represents energy lost as heat per cycle.

7. 5 A•m

Magnetic moment $M = m \times l$. $\rightarrow m = M / l = 0.5 / 0.10 = 5 \text{ A}\cdot\text{m}$.

8. $\sqrt{3}$ (≈ 1.732)

$B_V = B \sin\theta$, $B_H = B \cos\theta$. Ratio $B_H/B_V = \cot(30^\circ) = \sqrt{3}$.

9. 8000 A/m

$B = \mu_0(H + M)$ where $H = nI = 1000 \times 2 = 2000 \text{ A/m}$.

$B/\mu_0 = (4\pi \times 10^{-3}) / (4\pi \times 10^{-7}) = 10^4 = 10,000$.

$M = 10,000 - 2,000 = 8000 \text{ A/m}$.

10. $\chi = 0.5$ (Paramagnetic)

$\mu_r = 1 + \chi \rightarrow \chi = 1.5 - 1 = 0.5$. Since χ is positive and small, it is Paramagnetic.

11. Retentivity: The value of B when H is reduced to zero. **Coercivity:** The reverse field H needed to make B zero. Permanent magnets need high retentivity and high coercivity. Electromagnets need high retentivity but low coercivity.

12. Paramagnetic: Weakly attracted towards stronger parts of field (e.g., Al, Pt).
Ferromagnetic: Strongly attracted towards stronger parts of field (e.g., Fe, Ni).

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